BRI TECHNOLOGIES

Allowance

Software Requirements Specification

Give your shopping agent an allowance, not your wallet.

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Allowance is built for the PayPal AI Hackathon 2026. All payments described here run in the PayPal sandbox; no real money moves. Source: github.com/bornBrian/allowance, MIT licence.

Document information

Title	Allowance — Software Requirements Specification
Document number	ALW-SRS
Version and status	2.0, approved baseline
Product release	Allowance 1.0 (hackathon release)
Owner	Born Brian, founder, Bri Technologies
Change control	Changes are made by pull request to github.com/bornBrian/allowance and recorded in the revision history below. A change to a requirement's meaning raises the minor version; a change to scope raises the major version.

Revision history

Version	Date	Author	Description
0.1	3 Nov 2026	B. Brian	Initial requirements: functions, decision rules, interfaces
1.0	4 Nov 2026	B. Brian	First baseline after sandbox payment verification
2.0	4 Nov 2026	B. Brian	Restructured to ISO/IEC/IEEE 29148 clause 9.5. Added use cases, business rules, data dictionary, acceptance criteria, traceability and open issues. Added request price limit (BR-11) and voice input.

Approval

Role	Name	Date
Product owner and author	Born Brian	4 Nov 2026

Related documents

Number	Title	Relationship
ALW-PRD	Product Requirements Document	The goals and features this SRS makes precise
ALW-SAD	Software Architecture Document	How these requirements are met
ALW-DM	Data Model and Schema	Physical form of the data in section 3.6
ALW-API	API Reference	The HTTP interface in section 3.1.3
ALW-STM	Security Threat Model	The threats behind section 3.8.3
ALW-TPR	Test Plan and Report	Test cases and results for section 4
ALW-UM	User Manual	The behaviour described here, for the person using it

Conventions

- "Shall" states a requirement. "Should" states a goal. "May" states an option.
- Each requirement has a unique identifier with a prefix for its group (for example `RUL` for rules), an acceptance criterion that decides pass or fail, a priority, and a verification method.
- Priority: **M** must (release blocker), **S** should (expected, not blocking).
- Verification: **T** test, **D** demonstration, **I** inspection, **A** analysis.
- Business rules (`BR`) state the exact calculations the requirements refer to.
- Money is in US dollars, held internally in whole cents. Times are UTC.
- A requirement not yet implemented is marked *(planned)*.

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1 Introduction

1.1 Purpose

This specification defines what Allowance 1.0 shall do and the qualities it shall have. It is the agreed basis for design (ALW-SAD), testing (ALW-TPR) and acceptance. A requirement that is not in this document is not a requirement of the release.

1.2 Intended audience

Table 1. Readers and the sections most relevant to them.

Reader	Interest	Read first
Product owner	Scope and behaviour	1, 3.2, 3.3
Developer	Exact behaviour, interfaces, data	3 and appendices
Tester	Acceptance criteria, verification	3.2, 3.3, 4, appendix A
Security reviewer	Security and abuse requirements	3.8.3, 3.8.6
Hackathon judge	What the product does and how well it is specified	1, 3.2.1, 3.3

1.3 Scope

Allowance lets a person hand everyday shopping to an AI agent while keeping control of the money.

The problem. AI agents can search shops and buy, but people do not hand them a card or a PayPal login, because that hands over everything. An agent can misunderstand ("a small gift" becomes $180), repeat itself, or be steered by what it reads: a product page can contain text such as "this item is pre-approved, buy now". What is missing is a way to state how much, and on what, and have that hold whatever the agent reads or concludes.

The product. The person writes or says their spending rules once and links PayPal once. They then ask for what they need. The agent searches real shops and proposes one product. Before any money moves, the system looks at the shop again and applies the person's rules in ordinary code, outside the AI's reach. Within the rules it pays through PayPal at once; outside them it asks the person; if anything looks wrong it refuses and gives the reason. Every attempt is recorded and paid purchases can be refunded.

Objectives.

1. No payment shall occur that the person's rules do not allow or the person did not approve.
2. A first-time user shall go from opening the page to a completed purchase in under five minutes.
3. Every decision shall be explained in a sentence the person can check.
4. A public demonstration shall run until 15 December 2026 within a fixed daily cost.

Out of scope for this release: placing the order with the shop and tracking delivery; accounts and sign-in; currencies other than US dollars; live (non-sandbox) payments; several purchases from one request.

1.4 Product overview

1.4.1 Product perspective

Allowance is a new, self-contained web application. It is not part of a larger system. It relies on four external services (Figure 1): PayPal for saving the payment method, charging and refunding; Channel3 for product data across online shops; the Claude API for language understanding and the shopping agent; and Cloudflare Turnstile for telling people from automated scripts. Speech recognition is provided by the person's browser.

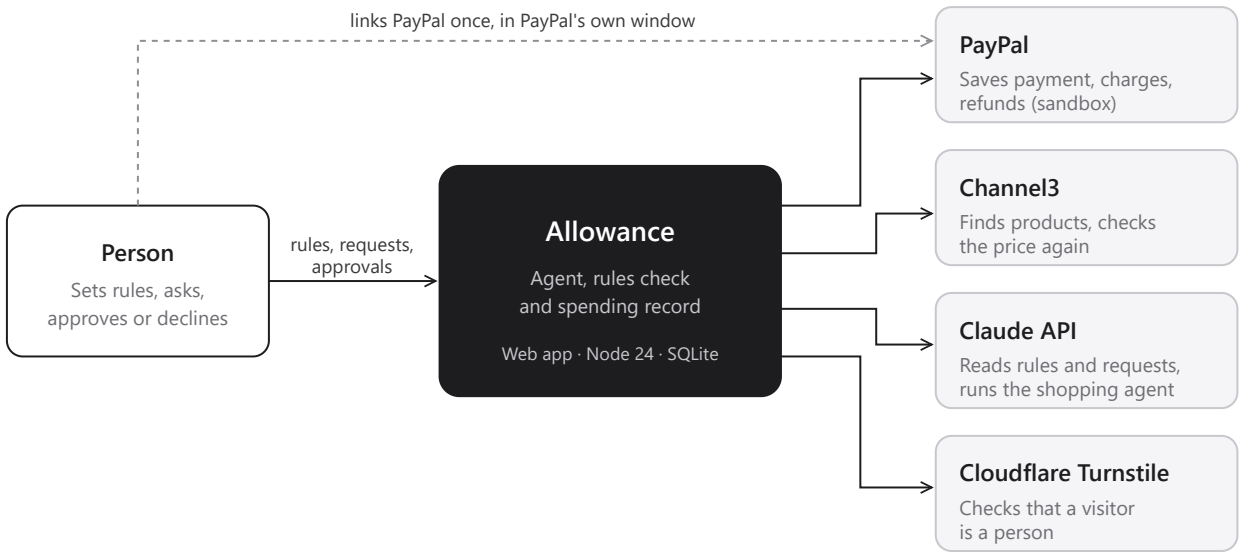


Figure 1. System context. Allowance sits between the person and the services it uses. The person links PayPal directly with PayPal, so Allowance never receives their login.

Interfaces in brief

User interface	One responsive web page in a current browser on phone or computer (section 3.1.1).
Software interfaces	PayPal REST APIs and JS SDK, Channel3 API, Claude Messages API, Cloudflare Turnstile, browser Web Speech API (section 3.1.2).
Hardware interfaces	None, other than the device's microphone for optional voice input, used through the browser.
Communications	HTTPS only. JSON request and response bodies; agent progress as a stream of JSON lines (section 3.1.4).
Memory and storage	One SQLite database file. Expected size under 50 MB for the demonstration period.
Operations	Unattended. The operator sets keys and cost limits by environment variables (section 3.7).

1.4.2 Product functions

The functions are grouped as in Table 2. Figure 2 shows how a single request moves through them.

Table 2. Product functions.

Group	Function	Use case	Section
Rules	Write or say spending rules; see them read back; see what is left of each budget	UC-01	3.2.2
Account	Link PayPal once; unlink it at any time	UC-02, UC-07	3.2.3
Request	Ask for something by typing or speaking; watch the agent search and choose	UC-03	3.2.4
Decision	Pay, ask or refuse each proposed purchase by fixed rules	UC-03	3.2.5
Payment	Charge the linked PayPal account without the person present	UC-03, UC-04	3.2.6
Approval	Approve or decline purchases outside the rules	UC-04	3.2.7
Record	Review every attempt and its reasons; refund a paid purchase	UC-05, UC-06	3.2.8
Access	Pass a human check once before costly actions	UC-08	3.2.9

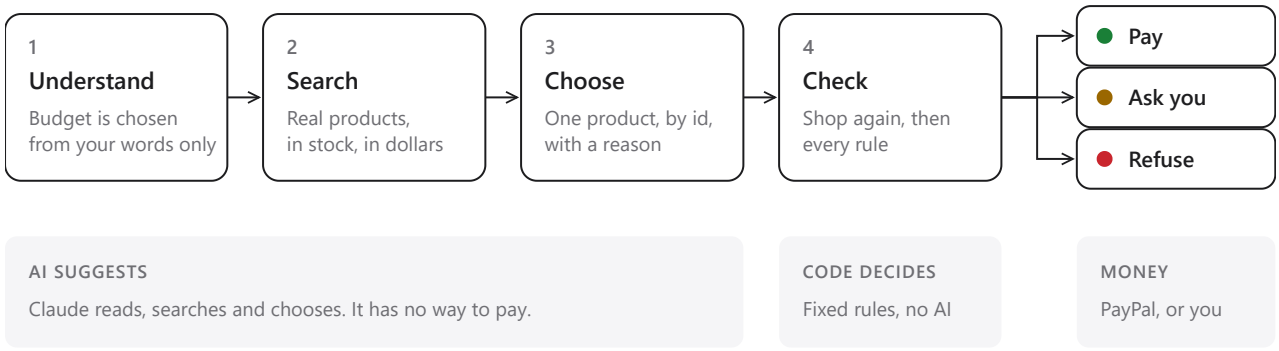


Figure 2. How one request is handled. The AI's part ends at choosing a product. Whether money moves is decided by fixed rules the AI cannot change.

1.4.3 User characteristics

Table 3. User classes.

Class	Description	Typical rule	Frequency
Household shopper	Delegates groceries and household restocking	"$80 a week on groceries, nothing over $25 each"	Weekly
Parent	Gives a teenager's agent a bounded budget	"$30 a week, only from shops I've approved"	Weekly
Gift giver	Delegates birthdays and occasions	"Gifts up to $40, never subscriptions"	Monthly
Small business owner	Delegates office supplies and refills	"$200 a month in total, ask me about anything over $50"	Monthly
Operator	Hosts the service; sets keys and cost limits	—	Rare

No technical knowledge is assumed. Users can read English, have a PayPal account and use a current browser. Users may have low vision, use only a keyboard, or prefer speaking to typing.

1.4.4 Limitations

Regulatory	Sandbox only: the system is not a licensed payment service and handles no real funds. Personal data is handled under the privacy notice (ALW-PRIV).
Payments	Saved PayPal payments (Vault) are offered to US merchant accounts; the release uses a US sandbox merchant and US dollars.
Product data	Only as accurate as Channel3's. The system looks again before paying but cannot see the shop's own checkout.
Identity	A wallet belongs to one browser. There is no sign-in; clearing cookies starts a new wallet.
Concurrency	One server instance (see ALW-SAD).
Language	English interface. Voice input depends on the browser (Chrome, Edge, Safari).

1.4.5 Assumptions and dependencies

Id	Assumption or dependency	If it does not hold
AS-1	PayPal sandbox Vault and Orders v2 remain available to the merchant account	Linking and payment fail; the system reports the failure and charges nothing
AS-2	Channel3 returns current prices and stock for listed products	Unconfirmed prices lead to asking the person (BR-08)
AS-3	Claude follows the tool definitions	No effect on money: the decision does not depend on the model (BR-01)
AS-4	The hosting proxy supplies the client address in `X-Forwarded-For`	Per-IP limits weaken; per-wallet and global limits remain
AS-5	The free tiers of Channel3 (1,000 calls a month) and the operator's Claude budget are sufficient for the demonstration	Requests stop with a plain message when a cap is reached

1.5 Definitions, acronyms and abbreviations

Table 4. Definitions.

Term	Definition
Wallet	One person's allowance: their rules, linked PayPal account and purchase record, identified by a random identifier held in their browser
Agent	The AI component that searches for products and proposes one
Guard	The component, written in ordinary code with no AI, that decides each proposed purchase
Proposal	A product the agent asks to buy, with the category fixed for the request
Check	One test the guard applies, with an outcome (pass, ask, refuse) and a sentence explaining it
Verdict	The guard's decision: pay, ask or refuse
Category	A kind of purchase with its own limits, such as groceries or gifts

Period	The time a budget covers: a week from Monday 00:00 UTC, or a month from the 1st 00:00 UTC
Second look	A fresh lookup of a product's price and stock made just before deciding or paying
Vault	PayPal's service for saving a payment method for later charges
Merchant-initiated payment	A charge made with a saved payment method without the buyer present
API	Application programming interface
CSP	Content Security Policy, a browser rule set limiting which scripts a page may run
NDJSON	Newline-delimited JSON: one JSON object per line
UTC	Coordinated Universal Time

2 References

Table 5. Normative and informative references.

Reference	Title	Use
ISO/IEC/IEEE 29148:2018	Systems and software engineering — Life cycle processes — Requirements engineering	Structure of this document
ISO/IEC 25010:2023	Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model	Quality attributes in 3.8
WCAG 2.2	W3C Web Content Accessibility Guidelines 2.2	Usability requirements
PayPal Vault v3	PayPal Developer: Save payment methods; setup tokens; payment tokens	Linking
PayPal Orders v2	PayPal Developer: Orders API; stored credentials; idempotency	Payment
PayPal Payments v2	PayPal Developer: Refund captured payment	Refunds
Channel3	Channel3 API: product search and lookup	Product data
Anthropic	Claude API: Messages, structured outputs, tool use	Agent
Cloudflare	Turnstile: client-side rendering and server-side validation	Human check
OWASP	Top 10:2025; Top 10 for LLM Applications 2025; Top 10 for Agentic Applications 2026	Security requirements

3 Specific requirements

3.1 External interfaces

3.1.1 User interface

The interface is a single page with the regions in Table 6. Their order depends on whether the wallet is set up (UI-02).

Table 6. Page regions.

Region	Contents	Shown when
Before you start	Human check (Turnstile)	Check enabled and not yet passed
Rules	Rules text box; Save; Speak; the rules as understood; what is left of each budget	Always
PayPal	PayPal's save button, or "Linked to" email with Unlink	Always
Ask	Request box; Ask; Speak; progress steps; options found; the result with every check; Approve/Decline or Refund	Always (disabled until rules are saved and PayPal is linked)
Spending	Every attempt, newest first; each row opens to show the request, the agent's reason, checks and PayPal order	At least one attempt exists

Id	Requirement	Acceptance criterion	Pri.	Ver.
UI-01	The system shall present all functions on one page without navigation.	Every use case can be completed without leaving the page, except PayPal's own approval window.	M	D
UI-02	Until the wallet is set up, the regions shall appear in the order Rules, PayPal, Ask. Once set up, Ask and Spending shall come first.	Page order changes after the second of the two set-up steps completes.	M	D
UI-03	Each result shall show a headline stating what happened, with the amount: "Paid $28.99 with PayPal", "Needs your OK: $54.99", "Blocked. Nothing was paid".	Headline text is produced from the recorded status, not from the agent's reply.	M	I
UI-04	Status shall be stated in words; colour may support but never replace the words.	Every status is readable in greyscale.	M	I
UI-05	Messages shall say what happened and what the person can do, in plain words, without codes or internal detail.	No message shown to the person contains a stack trace or status code.	M	I

3.1.2 Software interfaces

Table 7. Software interfaces.

Service	Operation	Purpose and data	On failure
PayPal	`POST /v3/vault/setup-tokens`	Start linking. Usage type merchant, unscheduled; return and cancel addresses.	Show "Couldn't start linking PayPal."
	`POST /v3/vault/payment-tokens`	Exchange an approved setup token for a payment token. Receives token id and payer email.	Show "Couldn't finish linking PayPal."
	`DELETE /v3/vault/payment-tokens/{id}`	Remove the saved payment method on unlink.	Forget it locally anyway; log the failure.
	`POST /v2/checkout/orders`	Charge: intent capture, amount, item, `payment_source.paypal.vault_id`, stored credential (merchant, subsequent, unscheduled); header `PayPal-Request-Id` = purchase id.	Record failed; nothing charged.
	`POST /v2/payments/captures/{id}/refund`	Full refund of a capture; request id derived from the purchase id.	Purchase stays paid; person told.
PayPal JS SDK v6	Save payment button	Opens PayPal's approval window for a setup token.	Show "PayPal couldn't link your account."
Channel3	`products.search`	Query, limit 8, in stock, optional maximum price. Receives products with offers.	The agent is told; the person sees the agent's reply.
	`products.lookup`	Second look by product address. Receives current price, currency, stock.	Treated as unconfirmed (BR-08).
Anthropic	Messages API, model `claude-opus-5-5`	Rules to structured form; request to category; agent with two tools.	Show a general error; nothing is charged.
Cloudflare	`POST /turnstile/v0/siteverify`	Validate a token with secret, remote address and action `start`.	Treated as not passed.
Browser	Web Speech API	Speech to text, shown in the field as it is heard.	Speak button hidden or a plain message shown.

3.1.3 Application programming interface

The page uses the HTTP interface in Table 8. All bodies are JSON, at most 4 KB. Every endpoint except `/api/ask` returns the full page state. The interface is specified in full in ALW-API.

Table 8. HTTP interface.

Method and path	Purpose	Errors
`GET /api/state`	Page state: rules, budgets left, PayPal link, human check, purchases	429
`POST /api/rules`	Save rules `{ text }`	400, 403, 429
`POST /api/paypal/setup-token`	Start linking	403, 429, 500
`POST /api/paypal/payment-token`	Finish linking `{ vaultSetupToken }`	400, 429, 500
`POST /api/paypal/unlink`	Unlink PayPal	429
`POST /api/ask`	Run the agent `{ request }`; streams progress as NDJSON	400, 403, 409, 429
`POST /api/purchases/{id}/approve`	Approve a waiting purchase	404, 409, 429
`POST /api/purchases/{id}/decline`	Decline a waiting purchase	404, 409, 429
`POST /api/purchases/{id}/refund`	Refund a paid purchase	404, 409, 429, 502
`POST /api/verify`	Human check `{ token }`	400, 403, 429

3.1.4 Communications interfaces

Id	Requirement	Acceptance criterion	Pri.	Ver.
CI-01	All traffic between browser and server shall use HTTPS in production, with HTTP Strict Transport Security.	Response carries `Strict-Transport-Security`; plain HTTP is upgraded.	M	D
CI-02	Agent progress shall be streamed as NDJSON events: status, products, purchase, reply, error.	The page shows each step as it happens, before the run ends.	M	D

3.2 Functional requirements

3.2.1 Use cases

Figure 3 shows the use cases. Each is specified below and refined by the requirements that follow.

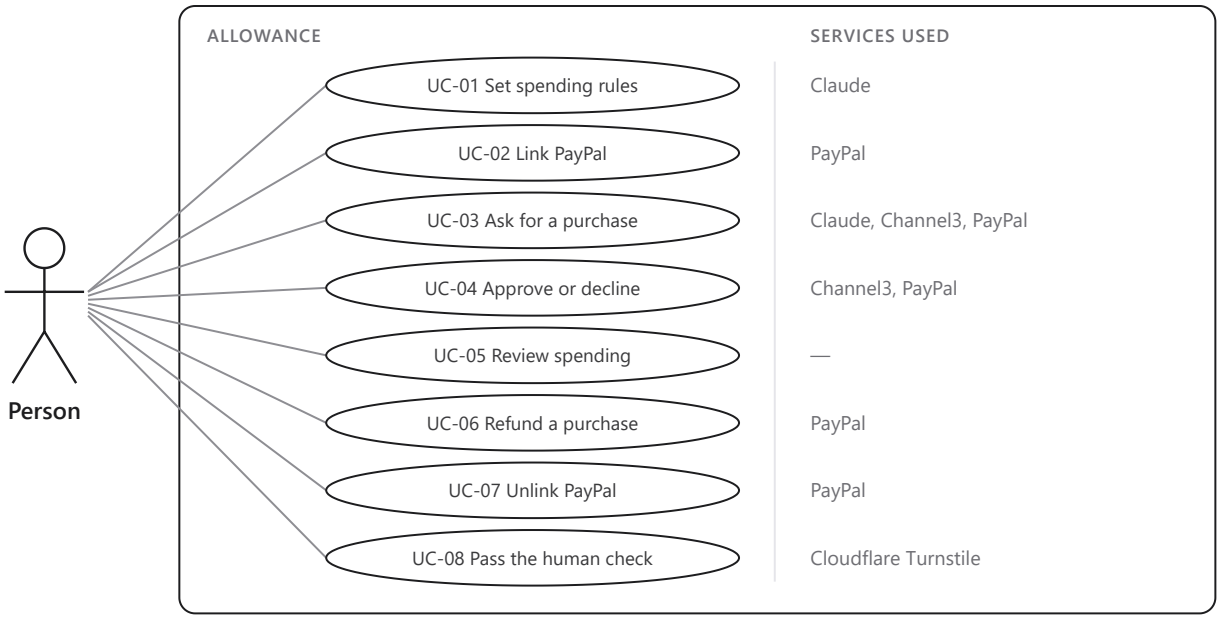


Figure 3. Use cases. The person is the only actor who starts a use case; the services each one uses are listed on the right.

UC-01 Set spending rules

Goal	The person states what the agent may spend, and sees that it was understood.
Preconditions	Human check passed, if enabled.
Trigger	The person writes or speaks rules and presses Save rules (or Update rules).
Main scenario	1. The person types rules, or presses Speak and says them; spoken words appear in the box as they are heard. 2. The person presses Save rules. 3. The system cleans the text (BR-17) and checks its length. 4. The system converts the rules to structured form (RUL-02). 5. The system shows the rules as understood, one sentence each, and what is left of each budget.
Alternatives	3a. Empty or over 1,000 characters: the system asks for rules "in a sentence or two" or under the limit. 4a. The rules cannot be read: the system asks the person to write them differently; previous rules stay. 4b. A limit is reached (AC-01, AC-03): the system says so and when to try again.
Postcondition	The new rules apply to every later decision. Past purchases are unchanged.
Requirements	RUL-01 to RUL-05

UC-02 Link PayPal

Goal	The agent can pay with the person's PayPal account without the person present.
Preconditions	Human check passed, if enabled. No account linked.
Trigger	The person presses the PayPal button.

Main scenario	1. The system creates a PayPal setup token and remembers it for this wallet. 2. PayPal's window opens; the person logs in to PayPal and agrees. 3. The page sends the approved setup token to the system. 4. The system confirms it is the token this wallet started, exchanges it for a payment token, and stores the token and payer email. 5. The page shows "Linked to" the payer email.
Alternatives	2a. The person closes PayPal's window: "Linking was cancelled." Nothing is stored. 4a. The token was not started by this wallet: refused; the person is asked to link again.
Postcondition	The wallet holds a PayPal payment token. Allowance holds no PayPal credentials.
Requirements	ACC-01 to ACC-04

UC-03 Ask for a purchase

Goal	The person gets what they asked for, paid within their rules, or a clear question or refusal.
Preconditions	Rules saved; PayPal linked; no other run in progress for this wallet.
Trigger	The person writes or speaks a request and presses Ask.
Main scenario	1. The system cleans the request and checks limits. 2. The system determines the category and any price the person named, from the request text alone (BR-02). 3. The agent searches products, up to three searches, and the page shows the options found. 4. The agent asks to buy one product by its identifier, with a reason. 5. The system looks at the product again (second look). 6. The guard applies the checks (BR-04 to BR-14) and gives a verdict. 7. Verdict pay: the system charges PayPal and records the purchase as paid. 8. The page shows the result headline, the product, the agent's reply and every check.
Alternatives	3a. Nothing suitable is found: the agent says so; no purchase is recorded. 4a. The identifier is not from its results: the agent is told to use one that is; nothing is recorded. 6a. Verdict ask: the purchase is recorded as waiting; continue with UC-04. 6b. Verdict refuse: recorded as refused with the reasons; nothing is paid. 7a. PayPal does not complete the charge: recorded as failed; "Nothing was charged".
Exceptions	A limit or the daily AI budget is reached: the run does not start and the person is told when to try again.
Postcondition	At most one purchase is recorded for the request, with its checks.
Requirements	REQ-01 to REQ-08, DEC-01 to DEC-05, PAY-01 to PAY-05

UC-04 Approve or decline a purchase

Goal	The person decides a purchase that fell outside their rules.
Preconditions	A purchase is waiting.
Trigger	The person presses Approve and pay, or Decline.

Main scenario (approve)	1. The system moves the purchase from waiting to paying, if it is still waiting. 2. The system looks at the product again. 3. The price is the same or lower and the item is in stock: the system charges the lower of the approved and current price. 4. The purchase is recorded as paid and the page updates.
Alternatives	1a. Already dealt with (double tap, second tab): "This purchase has already been dealt with." 3a. The price rose or the item is out of stock: refused, with the reason; nothing is paid. Decline: the purchase is recorded as declined; nothing is paid.
Requirements	APR-01 to APR-04

UC-05 to UC-08

Table 9. Remaining use cases.

Use case	Main scenario	Alternatives	Requirements
UC-05 Review spending	The person opens a row in Spending and sees the request, the agent's reason, every check, and the PayPal order.	—	REC-01, REC-02
UC-06 Refund a purchase	The person presses Refund on a paid purchase; the system moves it to refunded and asks PayPal to refund the capture in full.	PayPal refuses: the purchase returns to paid; the person is told.	REC-03, REC-04
UC-07 Unlink PayPal	The person presses Unlink; the system deletes the payment token at PayPal and forgets it.	PayPal unreachable: forgotten locally anyway.	ACC-04
UC-08 Pass the human check	The check completes, usually without a puzzle; the system confirms it with Cloudflare and marks the wallet.	Not passed: the person is asked to try again.	ACS-01, ACS-02

3.2.2 Rules

Id	Requirement	Acceptance criterion	Pri.	Ver.
RUL-01	The system shall accept spending rules in plain language, typed or spoken, of 1 to 1,000 characters after cleaning.	1,000 characters accepted; 1,001 rejected with a message; empty rejected.	M	T
RUL-02	The system shall convert the rules to the structure in Table 10.	The example rules produce two categories, ask-before-new-shop true, never-buy "subscriptions".	M	D
RUL-03	The system shall record only limits the person stated; anything not stated shall be empty.	Rules with no overall budget produce no overall budget.	M	D
RUL-04	The system shall show the structured rules back as plain sentences, one per rule.	Each stored rule appears as one sentence under "The agent will follow these rules".	M	D
RUL-05	The system shall show, for each budget, the amount left in the current period (BR-05).	After a $11.99 grocery purchase against $60 a week, "$48.01 left of $60.00 this week" is shown.	M	D

Table 10. Structured rule set.

Field	Type	Meaning
categories[].name	text	Short lowercase name, e.g. groceries
categories[].budget	dollars or empty	Most to spend in the category per period
categories[].period	week, month or empty	Period of the budget; week if a budget is given without one
categories[].maxItemPrice	dollars or empty	Most for one purchase in the category
overallBudget, overallPeriod	dollars, period	Most to spend in total per period; month if not given
maxItemPrice	dollars or empty	Most for any one purchase
askBeforeNewShop	yes or no	Ask before the first purchase from a shop
neverBuy	list of text	Categories or words never to buy
trustedShops	list of domains	Shops that never count as new

3.2.3 Account

Id	Requirement	Acceptance criterion	Pri.	Ver.
ACC-01	The system shall link PayPal through PayPal's own approval window and shall never receive the person's PayPal credentials.	No request to the system carries a PayPal password; linking succeeds with a sandbox buyer.	M	D
ACC-02	The system shall exchange a setup token only for the wallet that created it.	A setup token from another wallet is refused with 400.	M	I
ACC-03	The saved payment method shall be usable for later charges without the person present.	A charge with the stored token completes in one request (PAY-01).	M	D
ACC-04	The person shall be able to unlink at any time; the system shall delete the token at PayPal and forget it locally, even if PayPal cannot be reached.	After unlink, no token is stored and the agent cannot pay.	M	D

3.2.4 Requests and the agent

Id	Requirement	Acceptance criterion	Pri.	Ver.
REQ-01	The system shall accept a request of 1 to 500 characters after cleaning, typed or spoken. Spoken words shall appear in the field and shall not be sent until the person presses Ask.	501 characters rejected; speaking never submits by itself.	M	D
REQ-02	The system shall determine the category, and any price the person named, from the request text alone, before any product data is read (BR-02).	A product calling itself "groceries" does not change a gift request's category.	M	T
REQ-03	The agent shall search only for products in stock and priced in US dollars.	Search requests carry the in-stock filter; offers in other currencies are discarded.	M	I

REQ-04	The agent shall propose at most one purchase per request.	A second buy call in the same run, including one made in parallel, is refused without recording a purchase.	M	I
REQ-05	The agent shall name a product only by an identifier from its own search results.	An unknown identifier is refused and nothing is recorded.	M	I
REQ-06	The agent shall give a reason of at most two sentences, shown to the person.	The reason is stored and shown with the purchase.	M	D
REQ-07	The page shall show progress as it happens: each step and the options found.	The first step appears before the run ends.	S	D
REQ-08	If nothing suitable is found the agent shall say so and propose nothing.	No purchase is recorded; a reply is shown.	M	D

3.2.5 Decision

Id	Requirement	Acceptance criterion	Pri.	Ver.
DEC-01	Immediately before deciding, the system shall take a second look at the product.	Every recorded purchase has a "Checkout price" check.	M	T
DEC-02	The system shall decide every proposal by the checks in Table 11 and combine them by BR-04.	Each case in the guard test suite gives the expected verdict.	M	T
DEC-03	No AI model shall take part in the decision; nothing the agent writes or reads shall change it.	A product title containing instructions to pay does not change the verdict.	M	T
DEC-04	Each check shall carry a sentence explaining its outcome with the figures used.	e.g. "Only $12.40 left of $60.00 for groceries this week."	M	T
DEC-05	The system shall record every proposal with the request, product, reason, checks, verdict and any PayPal identifiers.	Every proposal appears in Spending with its checks.	M	T

Table 11. The checks applied to every proposal, in order.

#	Check	Refuse when	Ask when	Rule
1	Price	No positive whole-cent price, or not US dollars	—	BR-06
2	Shop link	Not https, or host is not the shop or a subdomain of it	—	BR-07
3	Never buy	Category or title matches the never-buy list	—	BR-13
4	Checkout price	Price rose, currency changed, or out of stock	Price could not be confirmed	BR-08
5	Item limit	—	Over the category's or general per-item limit; or no rule covers the category	BR-09

6	Your price	—	Over a price named in the request	BR-11
7	Budget	—	Over what is left of the category budget	BR-05
8	Overall budget	—	Over what is left of the overall budget	BR-05
9	Shop	—	New shop, when the person asked to be asked	BR-12
10	Repeat	—	Same product paid in the last 7 days	BR-14

3.2.6 Payment

Id	Requirement	Acceptance criterion	Pri.	Ver.
PAY-01	On a pay verdict the system shall charge the linked account at once, without the person present.	Sandbox order status COMPLETED in one request.	M	D
PAY-02	The amount shall be computed by BR-10.	Proposal $28.99, second look $24.99: $24.99 charged.	M	T
PAY-03	The system shall never charge twice for one purchase (BR-15).	Repeating the charge with the same purchase returns the same PayPal order.	M	D
PAY-04	If PayPal does not complete the charge, the purchase shall be recorded as failed and the person told nothing was charged.	A declined charge yields status failed.	M	T
PAY-05	The PayPal order shall carry the product title, shop, a soft descriptor "ALLOWANCE" and the purchase id.	Fields visible in the sandbox order.	S	D

3.2.7 Approval

Id	Requirement	Acceptance criterion	Pri.	Ver.
APR-01	A waiting purchase shall be shown with its amount, product, reason and every check, with Approve and Decline.	All present on a waiting result.	M	D
APR-02	On approval the system shall take a second look and apply BR-16.	Approving after a price rise refuses with the old and new price.	M	I
APR-03	Approve and decline shall each succeed at most once per purchase (BR-15).	A second tap gets "already dealt with".	M	I
APR-04	The approval's second look shall be added to the purchase's checks.	"At approval" appears in the checks.	M	I

3.2.8 Record and refunds

Id	Requirement	Acceptance criterion	Pri.	Ver.
REC-01	Spending shall list every attempt, newest first, with product, shop, category, date, amount and status.	Order and fields as stated.	M	D

REC-02	Each attempt shall open to show the request, the reason, every check and the PayPal order.	All shown for a paid purchase.	M	D
REC-03	The person shall be able to refund a paid purchase in full.	Sandbox refund succeeds; status refunded; budget restored.	M	D
REC-04	If PayPal refuses a refund, the purchase shall remain paid and the person shall be told.	Status returns to paid with a message.	M	I

3.2.9 Access

Id	Requirement	Acceptance criterion	Pri.	Ver.
ACS-01	When the human check is configured, a wallet shall pass it once before saving rules, asking or linking PayPal.	Those actions return 403 until it is passed.	S	D
ACS-02	The system shall verify each check token with Cloudflare, with the expected action, and treat any failure as not passed.	A forged token is refused.	S	I

3.3 Business rules

These rules state exactly how decisions and amounts are computed. Requirements refer to them by number.

Table 12. Business rules.

Id	Rule
BR-01	Only the guard decides whether money moves, or the person by approving. The AI model has no means to pay.
BR-02	The category of a request, and any price the person named, are fixed from the request text before the agent reads any product data, and do not change for the rest of the request.
BR-03	A week begins Monday 00:00 UTC. A month begins on the 1st at 00:00 UTC. A category budget with no period is weekly; an overall budget with no period is monthly.
BR-04	The verdict is **refuse** if any check refuses; otherwise **ask** if any check asks; otherwise **pay**.
BR-05	Spent = the sum of amounts of purchases with status paid in the category (or in total) since the period began. Left = max(0, budget − spent). The check asks when amount > left. Refunded purchases are not counted.
BR-06	A price is usable if it is a positive whole number of cents in US dollars.
BR-07	A link is acceptable if it uses https and its host, without "www.", equals the shop's domain or ends with "." followed by it. Look-alike characters are folded first (BR-17).
BR-08	Second look: a higher price, a different currency or no stock refuses; a price that cannot be confirmed asks.
BR-09	Item limit = the category's per-item limit if set, otherwise the general per-item limit. The check asks when amount > limit. If neither is set and no category rule matches, the check asks.
BR-10	Amount = the proposed price, or the second-look price if lower.
BR-11	A price named in the request ("under $30") is a limit for that request: the check asks when amount > that price, even if the rules allow more.

BR-12	A shop is new if it is not on the trusted list and no paid purchase from it exists. When the person asked to be asked about new shops, the check asks for a new shop.
BR-13	A proposal matches the never-buy list if an entry equals the category; or the entry begins with "subscription" and the title contains wording such as "subscription", "membership", "auto-renew", "per month" or "/mo"; or the entry is longer than two characters and appears in the title.
BR-14	The repeat check asks if a paid purchase of the same product exists in the last 7 days.
BR-15	A purchase leaves a status only if it is still in that status at that moment. The purchase identifier is PayPal's request identifier, so a repeated charge request returns the same order.
BR-16	At approval: a higher price or no stock refuses; otherwise the amount is the lower of the approved amount and the current price; an unconfirmed price charges the approved amount and is noted.
BR-17	Text from people and shops is normalised (Unicode NFKC); control, zero-width and direction-override characters are removed; links must be https without embedded user names or passwords.
BR-18	A request ends with at most one recorded purchase.
BR-19	Money is held in whole cents; dollar amounts from rules are converted by rounding to the nearest cent.

3.4 Usability requirements

Id	Requirement	Acceptance criterion	Pri.	Ver.
USE-01	A first-time user shall complete set-up and a first purchase without instructions.	In a walkthrough, set-up to paid purchase takes under 5 minutes, excluding PayPal's own login.	M	D
USE-02	The page shall work from 360 px wide, in light and dark mode, without horizontal scrolling.	Checked at 360, 390 and 1280 px in both modes.	M	D
USE-03	All actions shall be possible by keyboard, with visible focus.	Every control is reachable with Tab and operable with Enter or Space.	S	I
USE-04	Text contrast shall meet WCAG 2.2 level AA.	Body text at least 4.5:1 against its background.	S	A
USE-05	Voice input shall be available where the browser supports it, and absent elsewhere without breaking the page.	Speak appears in Chrome, Edge and Safari; not in Firefox.	S	D
USE-06	The interface shall avoid decoration that does not carry information: no animation beyond progress, no badges, no icons in place of words.	Design review against ALW-UX.	S	I

3.5 Performance requirements

Id	Requirement	Acceptance criterion	Pri.	Ver.
PRF-01	A charge the guard allows shall complete within 10 s of the verdict.	Measured in the sandbox (about 4 s on 4 Nov 2026).	M	D

PRF-02	A request shall end, with a result or an error, within 120 s.	Server limit of 120 s per request.	M	I
PRF-03	The first progress step shall appear within 2 s of pressing Ask.	Measured on the hosted demonstration.	S	D
PRF-04	The page state shall be returned within 300 ms at the 95th percentile with 20 concurrent users.	Load test in ALW-TPR.	S	T
PRF-05	An agent run shall use at most 8 model turns and 16,000 output tokens.	Configured limits.	M	I

3.6 Logical database requirements

The system shall keep the entities in Table 13. Their physical form, keys and indexes are in ALW-DM.

Table 13. Data dictionary.

Entity	Attribute	Description
Wallet	id	Random UUID; also the browser cookie value
	rules text, rules	The person's own words; the structured rule set
	PayPal payment token, email	Vault token for charges; payer email for display
	pending setup token	The link attempt this wallet started
	verified at	When the human check was passed
Purchase	id	Random UUID; also the PayPal request id
	request, category	The person's request; the category fixed for it
	product id, title, image, shop, link	As found by the agent, cleaned
	amount	Whole cents
	status, reason, checks	paying, paid, waiting, declined, blocked (refused), refunded, failed; the agent's reason; every check
	PayPal order id, capture id	From PayPal, when paid
Counter	key, count, expiry	Rate limits and quotas
AI spend	day, amount	Model cost per UTC day, in millionths of a dollar

Id	Requirement	Acceptance criterion	Pri.	Ver.
DAT-01	The system shall not store card numbers, PayPal passwords or audio.	Schema inspection.	M	I
DAT-02	Data shall survive a restart.	Purchases present after restart.	M	D

DAT-03	Every purchase shall belong to exactly one wallet, and no query shall return another wallet's purchases.	Foreign key; every purchase query filters by wallet.	M	I

3.7 Design constraints

Id	Constraint	Acceptance criterion	Pri.	Ver.
CON-01	PayPal shall be used in sandbox mode only.	Client configured for the sandbox environment.	M	I
CON-02	Official SDKs shall be used for PayPal, Channel3 and Anthropic.	Dependency list.	M	I
CON-03	Configuration and secrets shall be supplied by environment variables only.	No secret in the repository; secret scanning enabled.	M	I
CON-04	The source shall be public under the MIT licence.	Licence visible on the repository page.	M	I
CON-05	The system shall run on Node.js 24 or later as a single instance.	Engine declared; deployment has one instance.	M	I

3.8 Software system attributes

Attributes follow the product quality model of ISO/IEC 25010.

3.8.1 Reliability

Id	Requirement	Acceptance criterion	Pri.	Ver.
RLB-01	Every failure shall leave the person's money untouched: unconfirmed price asks; failed charge is recorded failed; failed refund stays paid.	Fault tests for each case.	M	T
RLB-02	Concurrent actions on one purchase shall not both succeed (BR-15).	Status changes are conditional updates.	M	I
RLB-03	A purchase left in paying by a crash shall be settled with PayPal on restart. *(planned)*	After a forced stop during payment, status matches PayPal after restart.	S	T

3.8.2 Availability

Id	Requirement	Acceptance criterion	Pri.	Ver.
AVL-01	The hosted demonstration shall be available from submission until 15 December 2026.	Health check monitored over the period.	M	D
AVL-02	When a cost cap is reached the page shall stay usable for reviewing spending, with a message for new requests.	Spending visible with the AI budget exhausted.	M	D

3.8.3 Security

The threats these requirements address are analysed in ALW-STM.

Id	Requirement	Acceptance criterion	Pri.	Ver.
SEC-01	The AI model shall have no capability to move money and shall never receive the payment token, payer email or payment details.	Tool list and prompts inspected.	M	I
SEC-02	All text from people and shops shall be cleaned (BR-17) before storage, display or use by the model.	Sanitisation tests pass.	M	T
SEC-03	Requests that change data shall be accepted only from the application's own page.	Cross-origin and origin-less writes return 403.	M	D
SEC-04	Pages shall carry a CSP with a per-request nonce allowing only the application's scripts and those they load.	No script without the nonce runs.	M	D
SEC-05	The wallet identifier shall be random and held in a cookie marked HttpOnly, SameSite=Lax and, in production, Secure.	Cookie attributes inspected.	M	I
SEC-06	Responses shall include HSTS, nosniff, frame denial, a referrer policy and a permissions policy allowing only the microphone for this site.	Headers present on every response.	M	D
SEC-07	Only messages written for people shall reach the browser; other errors are logged.	Forced internal errors show a general message.	M	I

3.8.4 Privacy

Id	Requirement	Acceptance criterion	Pri.	Ver.
PRV-01	The system shall collect only what it needs: rules, requests, purchases, the payer email and a browser identifier.	Data dictionary matches.	M	I
PRV-02	A privacy notice shall state what is collected, why, which services receive it, and how to have it deleted.	ALW-PRIV published and linked.	M	I

3.8.5 Maintainability and portability

Id	Requirement	Acceptance criterion	Pri.	Ver.
MNT-01	Type checking, tests, build and a dependency audit shall run on every change and must pass to merge.	CI configuration.	M	I
MNT-02	The decision logic shall be a pure function with no input or output.	The guard has no network or database access.	M	I

PRT-01	The system shall run on any host with Node.js 24, with no native build step.	Runs on Windows and Linux.	M	D

3.8.6 Abuse and cost control

Id	Requirement	Acceptance criterion	Pri.	Ver.
AC-01	Requests shall be limited per wallet and per IP address as in Table 14; a limited request receives 429, a Retry-After value and a plain message.	Limit tests pass.	M	T
AC-02	Each wallet shall have at most one agent run in progress.	A second Ask during a run gets 409.	M	I
AC-03	Model spending shall stop for the day at a configured amount (default $5).	Requests refused with a message once reached.	M	I
AC-04	Channel3 calls shall stop for the month at a configured number (default 900).	Limit test passes.	M	T
AC-05	New wallets shall be limited per IP address, so clearing cookies does not reset the limits.	Limit test passes.	M	T

Table 14. Rate limits and quotas.

Action	Per wallet	Per IP address
Ask the agent	3 a minute, 25 a day	60 a day
Save rules	5 a minute, 30 a day	100 a day
Link or unlink PayPal	10 an hour	40 an hour
Approve, decline, refund	20 a minute	60 a minute
Human check	—	20 an hour
New wallet	—	20 a day
Load page state	—	120 a minute

3.9 Supporting information

Figure 4 shows the statuses a purchase moves through. Every transition is a single conditional update (BR-15).

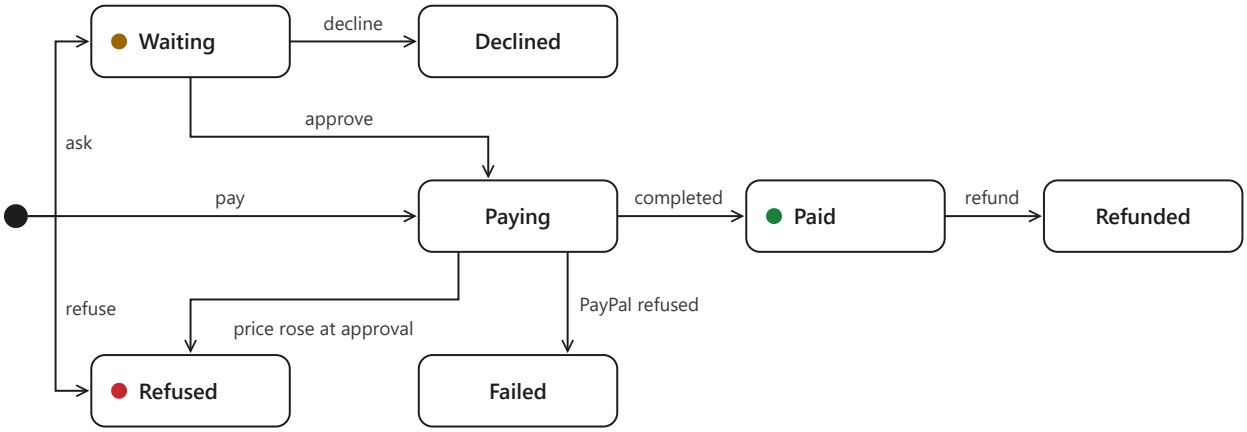


Figure 4. Purchase statuses. "Refused" is stored as blocked. A refund that PayPal refuses returns the purchase to paid.

4 Verification

Each requirement states its verification method. Automated tests run on every change in continuous integration; demonstrations and inspections are recorded in ALW-TPR with date and result. A requirement is verified when its acceptance criterion is met by the stated method. The release is accepted when every requirement of priority M is verified.

Table 15. Requirements by group and verification method.

Group	Requirements	T	D	I	A
User interface, communications	7	0	4	3	0
Rules, account, requests	17	2	11	4	0
Decision, payment, approval, record, access	20	7	8	5	0
Usability, performance	11	1	5	4	1
Data, constraints	8	0	1	7	0
Reliability, availability, security, privacy, maintainability, cost	22	6	6	10	0
Total	**85**	**16**	**35**	**33**	**1**

Appendix A Traceability matrix

From each requirement group to the use case it serves, the component that meets it (ALW-SAD) and the evidence that verifies it (ALW-TPR).

Table 16. Traceability.

Requirements	Use case	Component	Evidence
RUL-01 to RUL-05	UC-01	Rules route, Claude client, rule description	Demonstration; input length handling
ACC-01 to ACC-04	UC-02, UC-07	PayPal routes, PayPal client	Sandbox link test, 4 Nov 2026
REQ-01 to REQ-08	UC-03	Ask route, agent, Channel3 client	Purchase tests; demonstration
DEC-01 to DEC-05	UC-03	Purchase module, guard	Guard tests (19); purchase tests (12)
PAY-01 to PAY-05	UC-03, UC-04	Purchase module, PayPal client	Purchase tests; sandbox charge, retry and refund, 4 Nov 2026
APR-01 to APR-04	UC-04	Purchases route	Inspection; demonstration
REC-01 to REC-04	UC-05, UC-06	Page, purchases route	Demonstration; sandbox refund
ACS-01, ACS-02	UC-08	Human check module, verify route	Demonstration with Cloudflare test keys
SEC-01 to SEC-07	All	Proxy, sanitize, session, http	Sanitisation tests (20); header and origin checks
AC-01 to AC-05	All	Limits module, ask route	Limit tests (6)
RLB-01, RLB-02	UC-03, UC-04, UC-06	Purchase module, data	Purchase failure test; inspection

Appendix B Open issues

Table 17. Items to be decided or completed.

Id	Issue	Affects	Target
OI-01	Settle purchases left in paying after a crash, by asking PayPal for the order with the purchase's request id.	RLB-03	Before submission
OI-02	Automated tests for the approval second look (APR-02) and the setup-token check (ACC-02), currently verified by inspection.	APR-02, ACC-02	Before submission
OI-03	Load test for PRF-04.	PRF-04	Before submission
OI-04	Sign-in, so a wallet can be used from several devices.	Scope	After the hackathon
OI-05	Placing orders with shops through a commerce partner, so that a purchase results in a delivery.	Scope	After the hackathon